

NIDHAL DALHOUMI

Software Engineer | Full-Stack Development & Software Solutions

nidhal.dalhousmi@esprit.tn | [linkedin.com/in/nidhal-dalhousmi-1b4a721a6](https://www.linkedin.com/in/nidhal-dalhousmi-1b4a721a6) | nidhaldal.github.io | github.com/Nidhaldal

+ 216 92 233 120 | Tunis, Tunisia

EXPERIENCE

Medico International Services, End-of-Studies Project	2025 – 2025
- Engineered a full-stack medical web application using Angular and Django, translating user requirements into system design and scalable RESTful API architecture for clinical workflows. - Integrated real-time WebSocket features and a YOLO-based deep learning model for AI-assisted medical image analysis and decision support. - Designed backend architecture and application workflows to ensure efficient inter-module communication and scalable system operations. - Conducted requirement analysis and collaborated with end users to refine specifications, validate functionality, and align system behavior with operational needs. - Ensured system reliability through iterative testing, validation, and continuous performance and behavior improvements.	
IC CANADA, Software Engineering Intern	2024 – 2024
- Contributed to the design and development of a Business Intelligence and ERP platform aimed at optimizing business operations, reporting, and workflow management. - Analyzed business requirements and translated them into functional specifications, system workflows, and technical solutions. - Designed and implemented BI dashboards with filtering, aggregation, and reporting features to support data-driven decision-making. - Modeled business processes and optimized ERP workflows while designing relational database schemas and improving SQL performance. - Collaborated with business users to validate system outputs, ensure reporting accuracy, and align functionalities with operational needs.	

EDUCATION

Software Engineering, ESPRIT	2022 – 2025
Physics & Chemistry Preparatory Studies, IPEIT	2019 – 2022

CORE SKILLS

Full-Stack Development | Software Architecture & System Design | Backend Development | REST API Development | Database Design & SQL Optimization | Requirements Analysis | Functional Specifications | Business Process Modeling | Software Testing & Debugging | CI/CD & DevOps Practices | Agile / Scrum Development

LANGUAGES

English: B2 | French: B2 | Spanish: Basic

PROJECTS

DevOps & QA Automation Pipeline – ESPRIT	2025 – 2025
- Designed and implemented CI/CD pipelines using Jenkins, GitHub Actions, and Docker to automate build, test, and deployment workflows across development environments. - Integrated automated testing and code quality checks with SonarQube, Postman, and Swagger, and deployed containerized applications monitored via Prometheus and Grafana for reliability and performance tracking.	
Colocation & Carpooling Platform – ESPRIT, Spring Boot, Angular, REST APIs	
- Designed and developed a full-stack web platform enabling user matching workflows for housing and carpooling services. - Developed RESTful APIs using Spring Boot and built Angular interfaces to deliver a responsive and interactive user experience	

TECHNICAL STACK

Backend Development Django, Spring Boot, NestJS, REST APIs	Frontend Development Angular, React, TypeScript, JavaScript
Databases PostgreSQL, MySQL, SQL	DevOps Docker, Jenkins, GitHub Actions, Linux
Testing & Tools Postman, Swagger, UML	AI/Data Machine Learning, Deep Learning, YOLO, Power BI